

Calculus I - Derivatives Mini-Lecture

Four core ideas: instantaneous rate of change, power rule, product rule, and chain rule

1. Derivative as Instantaneous Rate of Change

Definition

A derivative measures how fast a quantity changes at a particular instant. If $y = f(x)$, then $f'(x)$ is the instantaneous rate of change of y with respect to x and the slope of the tangent line at that point.

Why it matters

Many quantities change continuously. Average speed describes an interval, while a derivative gives speed at one moment. The same idea appears in growth rates, marginal cost, velocity, acceleration, and optimization.

Concrete example

Suppose $s(t) = t^2$ meters. From $t = 2$ to $t = 2.1$, average velocity is $[s(2.1) - s(2)] / 0.1 = 4.1$ m/s. As the interval shrinks toward zero, this approaches 4 m/s, so $s'(2) = 4$ m/s.

Common misunderstanding

A derivative is not an average rate over a fixed interval. It is the limit of average rates as the interval shrinks to zero.

Useful comparison

Average rate of change uses two input values; instantaneous rate uses the derivative at one input and is the limit of those average rates.

2. Power Rule

Definition

For $f(x) = x^n$, the power rule gives $f'(x) = n x^{(n-1)}$: multiply by the exponent, then reduce the exponent by one. Constant coefficients remain in front.

$$\frac{d}{dx} [x^n] = n x^{(n-1)}$$

Why it matters

The power rule quickly differentiates polynomials and simple algebraic powers without returning to the limit definition each time.

Concrete example

If $f(x) = 7x^5$, then $f'(x) = 7(5x^4) = 35x^4$. For a polynomial, apply the rule term by term; constants differentiate to zero.

Common misunderstanding

For x^5 , the derivative is $5x^4$, not $5x^5$. Also, a nontrivial expression raised to a power, such as $(3x^2 + 1)^5$, needs the chain rule.

3. Product Rule

Definition

If $h(x) = u(x)v(x)$, then $h'(x) = u'(x)v(x) + u(x)v'(x)$. Each factor is differentiated once while the other is kept unchanged.

$$(uv)' = u'v + uv'$$

Why it matters

Differentiation does not distribute over multiplication. The product rule accounts for both factors changing at the same time.

Concrete example

Let $h(x) = x^2 \sin(x)$. Take $u(x) = x^2$ and $v(x) = \sin(x)$. Then $u'(x) = 2x$ and $v'(x) = \cos(x)$. Therefore $h'(x) = 2x \sin(x) + x^2 \cos(x)$.

Common misunderstanding

A frequent error is writing $(uv)' = u'v'$, which is generally false. For a constant times a function, such as $5x^3$, the constant can simply stay in front.

Useful comparison

Use the power rule for a single power such as x^6 and the product rule for two changing factors such as $x^2 \sin(x)$. A factor may still require another rule internally.

4. Chain Rule

Definition

The chain rule differentiates a composition. If $y = F(u)$ and $u = g(x)$, then $dy/dx = F'(u)g'(x)$: differentiate the outer function, leave the inner expression in place, then multiply by the inner derivative.

$$d/dx [F(g(x))] = F'(g(x)) g'(x)$$

Why it matters

Compositions appear in powers, trigonometric functions, exponentials, logarithms, and applied models. The chain rule tracks change through the inner and outer layers.

Concrete example

For $y = (2x + 3)^4$, the outer function is u^4 and the inner function is $u = 2x + 3$. The outer derivative is $4u^3$ and the inner derivative is 2, so $y' = 4(2x + 3)^3(2) = 8(2x + 3)^3$.

Common misunderstanding

A common mistake is differentiating only the outer layer. For $(2x + 3)^4$, writing $4(2x + 3)^3$ misses the required inner derivative, 2.

Useful comparison

The product rule handles multiplication; the chain rule handles nesting. For $x^2(3x + 1)^5$, the full expression is a product and one factor is also a composition.

Worked Example 1: Power Rule Across a Polynomial

Differentiate:

$$f(x) = x^4 - 3x^2 + 2x$$

Differentiate each term independently. The derivative of a sum or difference is the sum or difference of the derivatives.

Step 1: $d/dx[x^4] = 4x^3$

Step 2: $d/dx[-3x^2] = -3(2x) = -6x$

Step 3: $d/dx[2x] = 2$

Step 4: Combine the differentiated terms.

$$f'(x) = 4x^3 - 6x + 2$$

This example uses the power rule term by term. No product rule is needed because the coefficients -3 and 2 are constants, not changing functions of x .

Worked Example 2: Chain Rule for a Power of an Expression

Differentiate:

$$g(x) = (3x^2 + 1)^5$$

The fifth power is applied to the entire expression $3x^2 + 1$, so the function has an outer layer and an inner layer. That structure requires the chain rule.

Step 1: Identify the outer function. Let $u = 3x^2 + 1$. Then the outer function is $F(u) = u^5$.

Step 2: Differentiate the outer function: $F'(u) = 5u^4$.

Step 3: Identify and differentiate the inner function: $u = 3x^2 + 1$, so $u' = 6x$.

Step 4: Multiply the outer derivative by the inner derivative.

$$g'(x) = 5(3x^2 + 1)^4 (6x)$$

Step 5: Simplify the constant factors.

$$g'(x) = 30x(3x^2 + 1)^4$$

The factor $6x$ is essential because it is the derivative of the inner expression. Omitting it leaves the chain rule incomplete.

Connecting the Four Ideas

A derivative is an instantaneous rate of change. The power, product, and chain rules compute it for different structures. Before differentiating, identify whether the function is a power, product, composition, or combination.

Which Derivative Rule Should I Use?

Use the power rule for simple powers of x , such as x^7 , $-4x^3$, or $6x$. Multiply by the exponent and reduce the exponent by one.

Use the product rule when two differentiable expressions are multiplied. Differentiate one factor at a time, keep the other unchanged, and add the two terms.

Use the chain rule when a function is inside another function, such as $(3x^2 + 1)^5$, $\sin(x^2)$, or $e^{(4x)}$. Differentiate the outer layer and multiply by the inner derivative.

Some problems require more than one rule. In $x^2(3x + 1)^5$, the product rule organizes the full derivative, while $(3x + 1)^5$ also requires the chain rule. Choose rules from the expression's structure.

Fast structural check

Ask what the main operation is. A simple power points to the power rule, two changing factors to the product rule, and one function wrapped around another to the chain rule. If structures combine, apply the needed rules at each layer.

This handout supplies definitions, examples, rule distinctions, and common errors that an adaptive tutor can later use to generate diagnostics, hints, explanations, worked examples, or multiple-choice practice without a built-in quiz or answer key.